Ema Farshi, CS and Data Science

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EDUCATION

University Of Toronto | Honours Bachelor of Science Expected 2029
Computer Science and Data Science | GPA: 4.0/4.0
Relevant Coursework: Foundations of Computer Science I and II, Calculus with Proofs, Statistical Reasoning and Data Science

TECHNICAL SKILLS

Languages: Python, Java, R

Tools and Technologies: Git, GitHub, PyGame, Postgres SQL, Node.Js, Cursor

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Graphs, Trees, and Data Analysis

AI: Cursor, Spec Driven Development

EXPERIENCE

Teaching Assistant - Linear Algebra | University of Toronto Sept. 2026- Dec. 2026

- Support students in developing their understanding of linear algebra through in-class instruction and online discussion-board assistance
- Explain mathematical concepts and problem-solving to students with varying levels of experience
- Grade weekly coursework and provide feedback while assisting during lectures

PROJECTS

Recipe Recommendation System | Python, PyGame, Graphs, Trees CSC111

- Collaborated on an interactive recipe recommendation application using graph and tree based data structures to organize information and generate recommendations from user preferences
- Implemented search and filtering across ingredients dietary restrictions, excluded ingredients, and recipe categories to produce personalized results
- Built a graphical user interface in PyGame for collecting preferences and exploring recommended recipes

Pink Desktop Calculator | Python, tkinter, pytest

- Built a full-expression calculator supporting operator precedence, powers, percentages, degree-based trigonometry, and implicit multiplication without eval
- Applied spec-driven development to translate functional requirements into architecture, interface contracts, tasks, implementation, and automated tests
- Separated calculation, session-state and GUI layers into independently testable components

LEADERSHIP & INVOLVEMENT

Special Events Associate — Computer Science Students' Union (CSSU) May 2026 - Present

- Help organize major events for computer science students at the University of Toronto, including orientation, semi-formal events, and an undergraduate research conference

Seminars Executive — Undergraduate Research Students' Association (URSA) June 2026 - Present

- Help plan monthly research seminars and major events, including the How to Get Into Undergraduate Research (300+ attendees) and the Tri-Campus Undergraduate Research Conference